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PII: S0925-8388(18)32388-0

DOI: [10.1016/j.jallcom.2018.06.267](https://doi.org/10.1016/j.jallcom.2018.06.267)

Reference: JALCOM 46603

To appear in: *Journal of Alloys and Compounds*

Received Date: 1 April 2018

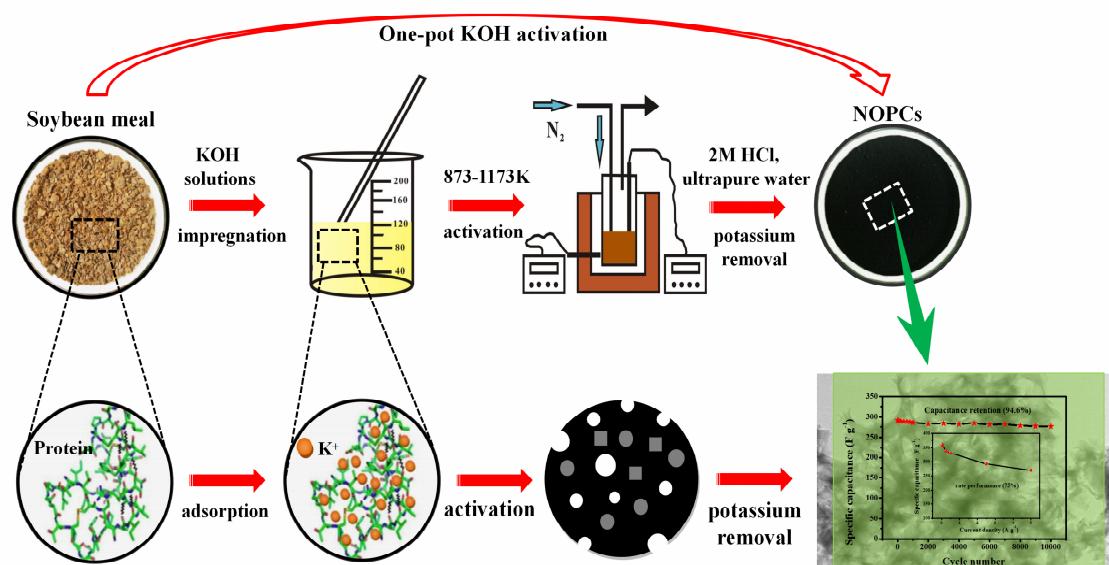
Revised Date: 21 June 2018

Accepted Date: 23 June 2018

Please cite this article as: H. Zhao, B. Xing, C. Zhang, G. Huang, Q. Liu, G. Yi, J. Jia, M. Ma, Z. Chen, C. Zhang, Efficient synthesis of nitrogen and oxygen co-doped hierarchical porous carbons derived from soybean meal for high-performance supercapacitors, *Journal of Alloys and Compounds* (2018), doi: 10.1016/j.jallcom.2018.06.267.

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Graphical Abstract



Efficient synthesis of nitrogen and oxygen co-doped hierarchical porous carbons derived from soybean meal for high-performance supercapacitors

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Abstract

Adopting an impregnation strategy, the homogeneous and efficient pore-creating are simultaneously achieved by one-pot KOH activation making full use of the swelling feature of hydrophilic soybean protein gels in alkaline solution. N, O co-doped porous carbons (NOPCs) have been fabricated using soybean meal as carbon, nitrogen and oxygen sources for an abundance of protein with a smaller amount of corrosive KOH. The as-obtained NOPCs exhibit advantageous characteristics for electrochemical energy storage including unique honeycomb-like morphology with interconnected carbon frameworks, high specific surface area, hierarchical porosity and controllable surface chemistry, which provides a large number of favorable nanochannels with suitable surface property for fast electrolyte diffusion and efficient electron transfer in the carbon networks. When tested as supercapacitor electrodes with symmetric configuration in 3 M KOH electrolyte, the NOPC-700 electrode prepared at 700 °C with a KOH/soybean meal mass ratio of 0.25 presents a high specific

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capacitance (330 F g^{-1} at 0.5 A g^{-1}), remarkable cycling stability (94.6% capacitance retention after 10000 cycles at 2.5 A g^{-1}), and superior rate capability (270 F g^{-1} at 5 A g^{-1}). Furthermore, the NOPC-700//NOPC-700 supercapacitor delivers a maximum energy density of 8.4 Wh kg^{-1} . The coordination between protein-rich soybean meal and one-pot KOH activation plays an important role in efficiently synthesizing porous carbons with robust energy storage performance for supercapacitors.

Keywords: Soybean meal, Swelling feature, One-pot KOH activation, N, O co-doped carbons, Hierarchical porosity, Supercapacitors

1. Introduction

Renewable solar and wind power has attracted great attention to meet the escalating energy needs. However, the majority of renewable energy is intermittent, which is preferably to be stored in energy devices [1]. Supercapacitors with advantages of high power density, rapid charge-discharge rates, wide operating temperature range, long lifetime as well as superior environmental friendliness have been employed as a promising candidate to fast store the excess electrical energy, and are widely applied in hybrid electric vehicles, stand-by power systems, and electronic devices [2-5]. Based on the mechanism of energy storage, supercapacitors can be categorized as electrical double-layer (EDL) capacitors and pseudocapacitors. Due to no electrochemical charge transfer kinetics constraint, EDL capacitors can obtain high power density ($>10 \text{ kW kg}^{-1}$), long cycling life (up to 100000 cycles), and good safety record [6-9]. Porous carbons have displayed great potential as efficient electrodes for EDL capacitors due to their high specific surface area for effective ion adsorption-desorption,

hierarchical porous structure for rapid ion diffusion, and good electrical conductivity for excellent cycling stability [10-12].

In general, porous carbons are prepared by template carbonization (using MgO [13-15], SiO₂ [16, 17], ZnO [18, 19], etc.), physical activation (treated in oxidizing atmospheres) and chemical activation (using KOH [20], KNO₃ [21], ZnCl₂ [22], etc.). It has been verified that template carbonization is beneficial to the preparation of porous carbons in both controlling the pore size and manipulating the morphology. However, homogeneous mesopores or macropores created by template carbonization can play the role of channels or ion-buffering reservoirs, respectively, leading to a lower EDL capacitance. Although physical activation is green, clean and non-corrosive, the method still needs relatively complicated apparatus and high energy consumption to obtain a well-developed porous structure. Compared with the two methods mentioned above, chemical activation based on potassium hydroxide is a typical and widely used method, by which activated carbons with significantly increased surface area, well-designed hierarchical porous structure and controllable heteroatom doping amount can be achieved by adjusting the activation condition and carbon source.

Modifying carbon frameworks with heteroatoms has been demonstrated to be favourable for enhancing the conductivity of carbon matrix, improving the wetting properties between electrodes and electrolytes as well as introducing additional pseudocapacitance via redox reactions [23, 24]. Among frequently-used heteroatoms such as nitrogen [25, 26], oxygen [27-29], boron [30, 31], sulfur [32-34], nitrogen is one of the most popular heteroatoms in carbon modification. Nitrogen-doped carbon materials have been successfully synthesized by treating carbonaceous precursors with ammonia gas [35], melamine [36-38], urea [39], etc. However, chemical nitrogen introduced by post treatment to the carbon products is easy to be lost during heat-treatment, resulting in a low nitrogen

content. Meanwhile, unexpected poor homogeneity also seriously affects the pore structure, surface chemistry, and electrochemical property of activated carbons. Fortunately, drawbacks of doping nitrogen by post treatment have been overcome by utilizing nitrogen-containing carbonaceous precursors, such as conductive polymers [40-43] and ionic liquids [44, 45]. Nitrogen could be preserved at a relatively high content by regulating the carbonization temperature and remain stable under a harsh working condition, but these materials are expensive and non-renewable [1]. Abundant nitrogen-containing biomass wastes, with little or no economic value, can be explored as low-cost carbonaceous precursors to prepare nitrogen-doped carbon materials. Furthermore, oxygen functional groups are commonly introduced by pyrolyzing various oxygen-rich carbonaceous precursors (biomass [46] and potassium humate [47]) or using strong oxidant [27, 48]. Therefore, nitrogen-containing biomass wastes could be one of the most attractive precursor candidates to prepare high added-value N, O co-doped carbon materials.

Recently, the advanced application of soybean-derived biomass wastes such as soybean root [49], *bradyrhizobium japonicum* [50] and soybean residue [51] in supercapacitor electrode materials has been studied, and these soybean-derived carbons exhibit outstanding electrochemical properties. Soybean meal, as the biomass waste of oil extraction, features cost-effective, abundant and renewable, thus scale-up production is highly promising. Importantly, soybean meal can provide essential carbon, nitrogen and oxygen groups during heat-treatment for heteroatom-doped carbon materials. More attractively, the combination between polar water molecules and hydrophilic protein gels by hydrogen bonding leads to a high swelling ratio. During expansion, more potassium ions enter into the interior of protein molecules, resulting a homogeneous activation instead of layer-by-layer KOH etching and a four-fold increase in the KOH efficiency compared with the

soybean root-derived hierarchical porous carbons [49].

In this work, we present an efficient strategy for the synthesis of N, O co-doped porous carbons (NOPCs) using protein-rich soybean meal as precursor via one-pot KOH activation. The efficient synthesis could be interpreted as follows: (i) great quantities of electrolyte ions could effectively absorb into the networks of protein molecules due to the hydrophilicity of soybean proteins, resulting in an extremely low dosage of KOH (KOH/soybean meal = 0.25). (ii) a facile impregnation treatment at ambient temperature takes full advantage of the swelling nature of soybean meal in aqueous solution. On the contrary, heat-treatment before activation can destroy the biological activity of soybean protein, leading to a significant increase in the amount of activating agent. (iii) the simple one-pot KOH activation combined with protein-rich biomass waste is very suitable for mass production. To the best of our knowledge, there is no relevant works reporting the efficient synthesis of N, O co-doped porous carbons by one-pot KOH activation derived from protein-rich soybean meal, which is the focus of this work. More importantly, the most suitable carbon electrode displays high specific capacitance (330 F g^{-1} at 0.5 A g^{-1}), good rate capability (75% capacitance retention from 0.05 to 5 A g^{-1}) and remarkable cycling stability (94.6% retention after 10000 cycles at 2.5 A g^{-1}), which presumably is closely related to the synergistic effect of high specific surface area, high mesoporous porosity, and the appreciable N and O doping amount.

2. Experimental

2.1. Preparation of NOPCs

Soybean meals were collected from north China (Shijiazhuang, Hebei). Scheme 1 describes the synthesis process of N, O co-doped porous carbons. The dried soybean meals were initially impregnated in $\sim 0.3 \text{ M}$ KOH solution at a soybean meal/KOH mass ratio of 4:1 overnight.

Subsequently, the mixtures were transferred to a vertical furnace for activation at 600-900 °C for 1 h in a N₂ atmosphere. The black solids were immersed in 2 M HCl solution overnight after natural cooling, then rinsed with ultrapure water several times until neutral pH was reached, and finally dried at 110 °C for 8 h. The as-prepared products were labeled as NOPC-600, NOPC-700, NOPC-800 and NOPC-900, respectively, where 600, 700, 800 and 900 refers to the activation temperature.

2.2. Characterization of NOPCs

The thermal decomposition behavior from 28 to 1000 °C was monitored by a NETZSCH STA 409 PC/PG thermal analyzer under a nitrogen flow of 30 ml min⁻¹. The morphology of NOPCs was observed by field emission scanning electron microscopy (FESEM, Quanta-250-FEG) with an energy dispersion spectroscopy analyzer and transmission electron microscope (TEM, JEM-2100). The porosity was examined at 77 K using an Autosorb-iQ-MP instrument (Quantachrome, USA). The products were degassed at 200 °C for 8 h prior to the measurement. Powder X-ray diffraction (XRD) patterns were recorded on a Rigaku SmartLab X-ray diffractometer with Cu K α radiation. Raman spectra were carried out using an inVia confocal Raman microscope (λ =633 nm). Fourier transformation infrared (FTIR) spectra were conducted on a Vector-22 spectrometer (Bruker, Germany) by the potassium bromide pellet technique. The elemental compositions were detected on a Vario-Macro-Cube elemental analyzer (Elementar, Germany). The surface functional groups of NOPCs were investigated by X-ray photoelectron spectroscopy (XPS) analysis on an Escalab 250Xi instrument (Thermo Fisher Scientific, USA).

2.3. Electrochemical measurements

In a two-electrode system, the working electrodes were fabricated by blending the products, carbon black, and polytetrafluoroethylene (PTFE) binder with a mass ratio of 85:10:5. The mixture was ground for 1 h, cut into pieces and pressed into disk-like electrodes (diameter = 13 mm) at 10

MPa. After drying at 120 °C for 4 h under vacuum, working electrodes were immersed in 3 M KOH aqueous solution for 1 h. Supercapacitors with symmetric configuration were assembled with two nearly identical electrodes (mass = 20 mg) that were separated by polypropylene membrane. That is, the mass of porous carbon (active material) is only ca.17 mg in single electrode. Three-electrode system was employed to confirm the existence of pseudocapacitance behaviors. Platinum foil, Hg/HgO, and 6 M KOH solution were adopted as counter electrode, reference electrode and electrolyte, respectively. The working electrode was prepared by loading a slurry consisted of 80 wt% active materials, 10 wt% of carbon black and 10 wt% PTFE in N-methyl-2-pyrrolidone on nickel foam, followed by vacuum drying at 120 °C overnight. Galvanostatic charge-discharge (GCD), cyclic voltammetry (CV), and cycling stability were characterized on a SCTS electrochemical workstation (Arbin, USA). Electrochemical impedance spectroscopy (EIS) was evaluated using a Parastat 2273 advance electrochemical system (Princeton Applied Research, USA). CV measurements were recorded in different potential ranges of 0.05-0.9 and -0.1-0.5 V for the two-electrode and three-electrode system, respectively. EIS measurements were operated in the frequency range from 10^{-3} Hz to 10^5 Hz with 10 mV amplitude. Cycling stability was recorded for 10000 cycles at a current density of 2.5 A g^{-1} .

In the two-electrode system, the specific capacitance of single electrode was calculated from the discharge curve from the formula: $C_s = 2I\Delta t / m\Delta V$, where I (A) is the discharge current, m (g) is the mass of active materials in single electrode, and ΔV (V) is the potential change within the discharge time Δt (s).

The energy density E (Wh kg^{-1}) and average power density P (W kg^{-1}) were defined by $E = 0.5C_{\text{cell}}(\Delta V)^2$ and $P = 3600E/\Delta t$, respectively, where C_{cell} (F g^{-1}) is the specific capacitance of a symmetrical two-electrode cell.

3. Results and discussion

After the sufficient KOH impregnation, the color of mixture shifting from bright yellow to dark

brown implies that KOH has successfully entered the networks of protein molecules (Scheme 1). The volume of mixture is expanded several-fold, resulting from the hydrophilicity of protein molecules. The unique property of soybean meal not only makes the activation more uniform instead of layer-by-layer etching for the common KOH activation, but also significantly reduces the dosage of KOH [53]. Subsequently, the impregnated soybean meals were directly activated to obtain the NOPCs. As presented in Table 1, the initial soybean meal has 8.71 wt% of N (37.73 wt% of O), which is higher than that of biomass wastes, such as willow catkins [54], typha angustifolia straws [55], and wheat straw [36], suggesting abundant nitrogen and oxygen functional groups in soybean meal.

Fig. 1 displays the thermogravimetric (TG) and derivative thermogravimetric (DTG) curves of raw soybean meal. Apparently, soybean meal goes through a three-step thermal decomposition process. The first stage takes place in the range of 41-191 °C centered at 109 °C with a mass loss of 9.2 wt%, due to the evaporation of free moisture and the dehydration of soybean meal. A weak shoulder peak at about 271 °C in DTG curve refers to the thermal decomposition of hemicellulose [54]. Furthermore, the rapid pyrolysis of proteins and cellulose between 191 and 563 °C mainly occurred at around 331 °C accompanying by an evident weight loss of 65.4% [54, 56]. Therefore, the precursors were impregnated in KOH solution at a mass ratio of 4 (soybean meal/KOH), which is determined by the char yield of ~25% at 600 °C and heat-treated up to the desired activation temperature (i.e. 600, 700, 800 and 900°C) according to the information disclosed from the TG and DTG curves.

The morphology of N, O co-doped porous carbons derived from soybean meal were characterized by FESEM and TEM. As shown in Fig. 2a-d, honeycomb-like structure with highly interconnected

frameworks can be observed easily for NOPC samples, which may be vital to the sufficient charge accumulation and good rate capability. In particular, numerous macropores with diameter of 50-600 nm and mesopores ranging from 2 to 50 nm disperse evenly on the surface of carbon materials. These pores can be served as ion-buffering reservoirs and channels to store more electrolyte ions and shorten the ion diffusion distance to the inner micropores, respectively [49]. It is clear that the NOPC-900 surface in Fig. 2d is etched by potassium hydroxide and covered by numerous small particles, which may be due to the excessive KOH activation at higher activation temperature [57]. Fig. 2e displays the TEM image of NOPC-700, from which the interconnected hierarchical frameworks can also be observed clearly. High-resolution TEM image of the boundary area depicts a long-range disordered structure with short-range curved graphitic lattices (Fig. 2f), which is in accordance to the result of SAED pattern in the inset with unclear rings. Elemental mapping images were performed to demonstrate the presence of heteroatoms in the as-obtained porous carbons (Fig. 2g). Apparently, nitrogen and oxygen atoms uniformly distribute in the carbon matrix. According to the relevant reports, the N at the center of the graphite plane could enhance the electrical conductivity of carbon materials, and the N at the edge of graphite plane would provide chemically active sites for pseudocapacitive interactions. In addition, the oxygen functional groups can not only improve the wetting properties of carbon materials owing to their proton donor property, also endow electrode materials with additional pseudocapacitance.

The XRD patterns of the as-resulted NOPCs are exhibited in Fig. 3a. The peaks at around 23.9° and 43.3° are interpreted as the (002) reflection of the turbostratic carbon and (100) diffraction of the graphitic carbon, respectively, revealing that all the samples have an amorphous structure accompanied by local graphitization. The presence of (100) peak also displays a higher interlayer

stacking extent that can effectively enhance the electrical conductivity of NOPCs [53]. As shown in Fig. 3b, the strong D bands (1340 cm^{-1}) in Raman spectra demonstrate the disordered structure in NOPCs, which is consistent with the XRD patterns. The NOPC-800 samples possess a lower I_D/I_G value of 1.12 compared with NOPC-600 (1.27), NOPC-700 (1.19) and NOPC-900 (1.20), which can offer good electrical conductivity for the relatively highly graphitized NOPC-800 [58]. Moreover, the higher I_D/I_G value of 1.20 for NOPC-900 results from the more defects created by strengthened KOH activation at higher activation temperature.

The elemental composition of the initial soybean meal is found to be 45.70 wt% C, 37.73 wt% O, and 8.71 wt% N. After heating at $700\text{ }^{\circ}\text{C}$, however, the elemental composition of NOPC-700 is converted to C (79.10 wt%), O (17.91 wt%), and N (2.26 wt%). As listed in Table 1, the bulk composition of NOPCs could be significantly affected by the activation temperature. Compared with NOPC-700 samples, a lower activation temperature brings about a higher N content for NOPC-600 (5.63 wt% N). On the contrary, the higher activation temperatures result in the lower N and O contents for both NOPC-800 (N 1.36 wt%, O 12.52 wt%) and NOPC-900 samples (N 1.34 wt%, O 8.52 wt%). Therefore, we still have difficulty in synthesizing the carbon material with high porosity and high heteroatom content at high temperature, because these pores are created after the gas volatilization.

XPS measurements were conducted to reveal the chemical identification of heteroatoms in the as-prepared NOPCs. The synchronized appearance of C 1s (285 eV), O 1s (532 eV) and N 1s (400 eV) peaks in the survey spectra demonstrates the successful synthesis of N, O co-doped carbon materials (Fig. 3c). The percentage of nitrogen in NOPC-600, NOPC-700, NOPC-800 and NOPC-900 is determined from the XPS survey spectra to be 4.95, 3.12, 1.07 and 0.47 at%,

respectively (see Table 1). Particularly, the N/C mass ratio of NOPC-700 calculated by elemental analysis is much lower than that measured by XPS, indicating that the N concentrations on the surface is significantly higher than that in the bulk materials. This is very important for NOPC-700 applied in supercapacitors, because only the heteroatoms on the surface of materials contribute to the pseudocapacitance and electrical conductivity [55]. As shown in Fig. 3d, the high-resolution N 1s spectra can be fitted by four component peaks corresponding to pyridinic nitrogen (N-6, 398.7 ± 0.3), pyrrolic or pyridonic nitrogen (N-5, 400.3 ± 0.3), quaternary nitrogen (N-Q, 401.3 ± 0.1), and pyridine nitrogen oxides (N-X, 402.6 ± 0.2). N-Q is the predominant nitrogen configuration on the surface of NOPC-800 samples, and its composition is about 0.49 at%, which is higher than the sum of the basic N-6 and N-5 species. N-Q may be advantageous for enhancing the electrical conductivity for NOPC-800. In comparison to NOPC-800, NOPCs prepared at lower temperatures possess higher contents of N-6 and N-5. With reference to the previous report, the N at the edge of graphite plane (N-6 and N-5) may be more active than N-Q for offering redox reaction sites on the surface of electrodes [59]. As presented in Fig. 3e, the O 1s XPS spectra of NOPCs can be divided into three different binding configurations, including C=O (531 ± 0.2), C-O (532.4 ± 0.3), and -COOH (533.9 ± 0.5). It is reported that basic oxygen groups (C=O and C-O) have exhibited good electrochemical redox activity to enhance the pseudocapacitance (see Table 2). In addition, the analysis from the deconvoluted C 1s spectra reveal that about 41 at% C atom is directly bonded to N and/or O atom in NOPC-700 (see Table S1). The high level of N and O functional groups is conducive to improving the wetting properties of materials to facilitate the accessibility of the electrolyte ions and inducing the redox reactions to contribute to pseudocapacitance.

The nature of doping for NOPCs was further confirmed by FTIR. The broad peak at 3425 cm^{-1} is

ascribed to the symmetric stretching vibration of N-H bond and/or O-H bond in hydroxyl (Fig. 3f). The sharp peak centered at 1584 cm^{-1} is corresponding to the in-plane bending vibration of N-H bond. The broad peak located at 1200 cm^{-1} can be assigned to the stretching vibration of C-O bond in free phenolic hydroxyl, and the accompanying O-H bond appears at 3610 cm^{-1} , which can be clearly seen in the spectrum of NOPC-700. These peaks become broader and weaker with increasing the activation temperature from 600 to 900 °C, which may be owing to the destruction of chemical bonds at high temperature. Furthermore, in terms of NOPC-700, the stretching vibration of C=O bond in the carboxyl groups occurs at 1791 cm^{-1} , and the accompanying symmetric stretching vibration of O-H appears at 3207 cm^{-1} , showing weak peaks. Therefore, the FTIR analyses further demonstrate the presence of nitrogen and oxygen functional groups in the obtained porous carbons.

N₂ adsorption-desorption isothermal analysis was performed to determine the porosity of NOPCs. All isotherms display the combination of type $\square$ and $\square$ characteristics with a weak type-H4 hysteresis loop, indicating the presence of micropores and mesopores (Fig. 4a). The obvious steep rise of N₂ uptake at high relative pressure (0.95-1) implies the retention of some slit-shaped macropores, which can be attributed to the remained honeycomb-like walls and stacking carbon particles [60]. The collective isotherms suggest that micro-, meso-, and macropores exist in NOPCs [58]. The pore size distribution is estimated by the conventional DFT method (Fig. 4b). The NOPCs have sharp peaks at 1.18, 1.41, and 1.85 nm as well as slightly sharp one at around 2.65 nm, indicating the coexistence of hierarchical micropores and mesopores. As listed in Table 3, the specific surface area increases from 633 to 2059 m² g⁻¹, as determined by the BET method. The void spaces would be generated from the gas evolution and occupied by the resultant potassium compounds, such as K₂CO₃ and K₂O at high temperature. After these compounds are removed with ultrapure water, the

number of micropores could be dramatically boosted [57]. High specific surface area can offer an ample electrode/electrolyte interface for ion or charge accumulation. Moreover, the total pore volume of $0.78 \text{ cm}^3 \text{ g}^{-1}$, the micropore volume of $0.38 \text{ cm}^3 \text{ g}^{-1}$, the mesopore volume of $0.40 \text{ cm}^3 \text{ g}^{-1}$, the mesopore volume/total pore volume ratio of 50.6%, and the average pore diameter of 2.66 nm can also be found in Table 3 for NOPC-700. As compared to the microporous carbon synthesized from willow catkins by KOH activation (which typically possesses pores of 2 nm or smaller), the present high mesoporous porosity is more favorable for rapid ion diffusion and superior rate performance [54]. In addition, the appropriate proportion of micropores guarantees the high specific surface area for NOPC-700 [58].

From the above discussion, we can conclude that the activation at 700°C made the initial soybean meal turn into the N, O co-doped hierarchical porous carbons with interconnected pores composed of micropores, mesopores and macropores, high specific surface area, high mesoporous porosity, localized graphitic structure, and appreciable N and O doping amount. The one-pot KOH activation provides an effective way to prepare well-controlled NOPCs. Therefore, the efficient synthesis of NOPCs from renewable biomass waste through one-pot KOH activation makes NOPC-700 a promising candidate for supercapacitor electrode materials.

CV and GCD techniques were conducted to evaluate the electrochemical performances of NOPCs using two-electrode system in 3 M KOH electrolyte. Fig. 5a shows the CV curves of NOPCs in symmetric supercapacitors at a scan rate of 1 mV s^{-1} . The CV curves of NOPC-800 and NOPC-900 in Fig. 5a exhibit roughly rectangular shape, indicating that EDL capacitance is the dominant contributor to the total capacitance based on attraction and separation of charges, while NOPC-600 and NOPC-700 present wide reversible humps due to the synthetic effects of EDL capacitance and

pseudocapacitance. The closed CV curve of NOPC-900 displays the minimum area, suggesting the lowest specific capacitance (Fig. 5a). Although the specific surface area of NOPC-900 is the largest among the four samples, a large number of micropores would limit ion diffusion into the electrode material. This ultimately leads to inefficient utilization of specific surface area. We also observe that NOPC-700 electrode still maintains the rectangular-shaped CV curves even at 40 mV s^{-1} , which benefits from the excellent electrical conductivity of NOPC-700 samples (Fig. 5d). In contrast, the CV curves of NOPC-600 electrode are obviously distorted with increasing the scan rate from 1 to 40 mV s^{-1} owing to the relatively large internal resistance caused by the limited ion incorporation into the active electrode material (Fig. 5a-d). Three-electrode configurations were assembled to further confirm the pseudocapacitance behavior of NOPC-700 electrode in 6 M KOH electrolyte. As seen from Fig. 5e, the obvious redox peaks are observed at 0.20-0.40 V (vs Hg/HgO) of the CV curves determined at low scan rates and ascribed to the pseudo-capacitive contribution of the surface N functional groups [55]. While there is no distinct redox peak appeared in the CV curves of NOPC-800 electrode (Fig. 5f), which may be due to the low concentration of the surface N functional groups.

The GCD tests were conducted at 0.05 A g^{-1} to evaluate the capacitive behavior of all samples (Fig. 6a). The NOPC-700 electrode shows the longest discharge time, whereas NOPC-600, NOPC-800 and NOPC-900 exhibit slightly reduced discharge time. This observation suggests that NOPC-700 electrode offers the largest capacitance (360 F g^{-1}), followed by NOPC-600 (340 F g^{-1}), NOPC-800 (280 F g^{-1}) and NOPC-900 (247 F g^{-1}), which is in accordance with CV analysis at 1 mV s^{-1} . The nearly symmetric charge-discharge curves of NOPC electrodes indicate the EDL mechanism dominates. The asymmetry part is caused by the pseudocapacitive behaviors of the N and O

functional groups on the surface. Additionally, for the charge-discharge curves of NOPC-700 electrode at a high current load of 5 A g^{-1} (Fig. 6b), the ohmic drop is rather limited compared with other electrodes owing to the rich mesopores and good conductivity of NOPC-700 sample. Rich mesopores can accelerate the ion transfer and reduce the inner resistance of electrodes [1]. It is also found in Table S2 that the specific capacitance for NOPC-700 electrode is 330 F g^{-1} at 0.5 A g^{-1} , which is much higher than NOPC-600 (295 F g^{-1}), NOPC-800 (256 F g^{-1}), and NOPC-900 electrodes (221 F g^{-1}). Evidently, the obtained specific capacitance of NOPC-700 electrode (330 F g^{-1} at 0.5 A g^{-1}) is quite advantageous in comparison to the most reported biomass-derived porous carbons under similar test conditions (see Table S3). For example, the specific capacitance of soybean meal-derived porous carbons by a staged activation process was 249 F g^{-1} at 0.5 A g^{-1} (Fig. S2-4 and Table S2), demonstrating the efficient synthesis of N, O co-doped porous carbons via one-pot KOH activation. According to the relevant reports, the high specific capacitance for NOPC-700 electrode may be owing to the collaborative effects of a high specific surface area that enhances the surface area accessibility for charge accumulation, a high mesoporous porosity that accelerates ion transportation and diffusion, and a high nitrogen and oxygen doping amount that changes the electron donor/acceptor characteristics of carbon. The charge transport and ion diffusion kinetics as well as the internal resistance of carbon materials were investigated by EIS measurements. Fig. 6c shows the Nyquist plot of NOPC-700 electrode measured from 10^{-3} Hz to 10^5 Hz in a two-electrode system. Notably, at low frequency, the Nyquist plot gives a steep curves nearly parallel to Z'' , indicating the fast ion diffusion in the nanochannels of NOPCs [55]. A small Warburg portion at middle frequency and a small semicircle (4.43Ω) at high frequency region corresponds to the fast ion diffusion into the carbon electrodes and a lower charge transfer resistance, respectively [61]. In addition, the intercept

at real axis (0.17Ω) indicates a low contact resistance and good conductivity, which is advantageous for excellent capacitance performance for NOPC-700 electrode.

Cycling performance of NOPC-700//NOPC-700 supercapacitor was evaluated by continuous GCD measurements at 2.5 A g^{-1} for 10000 cycles. The specific capacitance of NOPC-700 electrode as a function of cycle number based on the two-electrode system is provided in Fig. 6d. Clearly, the specific capacitance still remains 275.8 F g^{-1} (94.6% capacitance retention), which demonstrates a remarkable cycling stability and high reversibility. As shown in the inset of Fig. 6d, the rate performance at a range of current density from 0.05 to 5 A g^{-1} reports that the capacitance retention of NOPC-700 electrode is $\sim 75\%$. The significantly improved rate capability can be related with the hierarchical porous structure of NOPC-700 sample. Mesopores and macropores are able to furnish convenient transport and diffusion channels for electrolyte ions, while the micropores offer the intrinsic area for charge accommodation [60]. Furthermore, the NOPC-700//NOPC-700 supercapacitor delivers a maximum energy density of 8.4 Wh kg^{-1} , suggesting the promising application of soybean meal-derived nitrogen and oxygen co-doped porous carbons in supercapacitor electrode materials.

4. Conclusions

Based on the swelling feature of hydrophilic soybean protein gels in alkaline solution, efficient synthesis of nitrogen and oxygen co-doped hierarchical porous carbons has been achieved by one-pot KOH activation with an extremely low dosage of KOH, which is promising for mass production. Such favorable features of high specific surface area ($1175 \text{ m}^2 \text{ g}^{-1}$), high mesoporous porosity (50.6%), and appreciable N and O doping amount (3.1 and 11.2 wt%, respectively) endow the N, O co-doped porous carbons synthesized at 700°C (NOPC-700) with a high specific capacitance (330 F g^{-1})

g⁻¹ at 0.5 A g⁻¹), good rate capability (75% capacitance retention at 5 A g⁻¹), and remarkable cycling stability (94.6% retention after 10 000 cycles at 2.5 A g⁻¹). The symmetric supercapacitors in KOH electrolyte based on the NOPC-700 electrodes also exhibit a maximum energy density of 8.4 Wh kg⁻¹. These results demonstrate that the N, O co-doped porous carbons exhibited here could be a promising candidate for supercapacitor electrode. Enlarging the cell voltage helps to obtain a higher energy density without sacrificing the power density and cycle life. Further load of transition metal oxide on the basis of rich mesopores and good conductivity may be a feasible way to harvest high-quality and reliable soybean meal-derived electrode materials for high-performance supercapacitors.

Acknowledgements

This work was financial supported by the National Natural Science Foundation of China (U1361119), the Natural Science Foundation of Henan province (162300410115) and the research fund of Henan Key Laboratory for Green and Efficient Mining & Comprehensive Utilization of Mineral Resources (Henan Polytechnic University).

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Table 1 Elemental analysis, the deconvoluted N 1s peaks from XPS spectra, and their relative percentage of nitrogen functional groups in terms of NOPCs.

Table 2 Analysis of the deconvoluted O1s peaks from XPS spectra and their relative percentage of oxygen functional groups in terms of NOPCs.

Table 3 Textual parameters of NOPCs.

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Scheme 1. Schematic illustration for the fabrication processes of NOPCs.

Fig. 1. TG and DTG curves of raw soybean meal.

Fig. 2. (a-d) FESEM images of NOPC-600, NOPC-700, NOPC-800 and NOPC-900. (e) TEM image, (f) high-resolution TEM image, and (g) elemental mapping images based on the Fig. b of NOPC-700.

Fig. 3. (a) XRD patterns, (b) Raman spectra, (c) XPS survey spectra, high-resolution XPS spectra of the deconvoluted (d) N1s and (e) O1s peaks, and (f) FTIR spectra of NOPCs.

Fig. 4. (a) Nitrogen adsorption-desorption isotherms and (b) pore size distribution (inset: magnified 0.6-2 and 2-5 nm region) of NOPCs.

Fig. 5. Electrochemical performance of NOPC electrodes in a two-electrode system using 3 M KOH electrolyte: (a-d) Cyclic voltammograms for various scan rates; Cyclic voltammograms of (e) NOPC-700 and (f) NOPC-800 electrode for various scan rates measured in a three-electrode system using 6 M KOH electrolyte.

Fig. 6. (a) Charge-discharge curves of NOPC electrodes at a current density of 0.05 A g^{-1} ; (b) Charge-discharge curves of NOPC-700 electrode at various current densities; (c) Electrochemical impedance spectrum (inset: magnified high and middle frequency region) collected under the influence of an ac voltage of 10 mV; (d) Cycling stability at 2.5 A g^{-1} up to 10000 cycles (inset: specific capacitance as a function of current density).

Table 1

Carbon Materials	Elemental analysis (wt%)			Surface N (at%)	N-6 (398.7±0.3 eV)	N-5 (400.3±0.3 eV)	N-Q (401.3±0.1 eV)	N-X (402.6±0.2 eV)
	C	N	O					
Soybean meal	45.70	8.71	37.73					
NOPC-600	75.77	5.63	16.97	4.95	25.5% (1.26)	47.4% (2.35)	27.1% (1.34)	0
NOPC-700	79.10	2.26	17.91	3.12	31.0% (0.97)	37.2% (1.16)	19.8% (0.62)	12.0% (0.37)
NOPC-800	85.66	1.36	12.52	1.07	11.2% (0.12)	22.8% (0.24)	45.4% (0.49)	20.6% (0.22)
NOPC-900	89.67	1.34	8.52	0.47	48.8% (0.23)	13.1% (0.06)	26.8% (0.13)	11.3% (0.05)

Table 2

Carbon Materials	Surface O (at%)	C=O	C-O	O=C-O
		(531 ± 0.2 eV)	(532.4 ± 0.3 eV)	(533.9 ± 0.5 eV)
NOPC-600	10.85	25.0% (2.71)	35.0% (3.80)	40.0% (4.34)
NOPC-700	11.22	16.5% (1.86)	51.0% (5.72)	32.5% (3.64)
NOPC-800	9.67	11.3% (1.09)	54.9% (5.31)	33.8% (3.27)
NOPC-900	6.60	24.2% (1.60)	50.4% (3.32)	25.4% (1.68)

Table 3

Carbon Materials	$S_{\text{BET}}^{\text{a}}$ ($\text{m}^2 \text{g}^{-1}$)	V_{t}^{b} ($\text{cm}^3 \text{g}^{-1}$)	$V_{\text{mic}}^{\text{c}}$ ($\text{cm}^3 \text{g}^{-1}$)	$V_{\text{mes}}^{\text{d}}$ ($\text{cm}^3 \text{g}^{-1}$)	$V_{\text{mes}}/V_{\text{t}}$ (%)	D_{ap}^{e} (nm)
NOPC-600	633	0.48	0.19	0.29	60.1	3.03
NOPC-700	1175	0.78	0.38	0.39	50.6	2.66
NOPC-800	1700	0.98	0.60	0.38	39.0	2.31
NOPC-900	2059	1.34	0.72	0.62	46.2	2.60

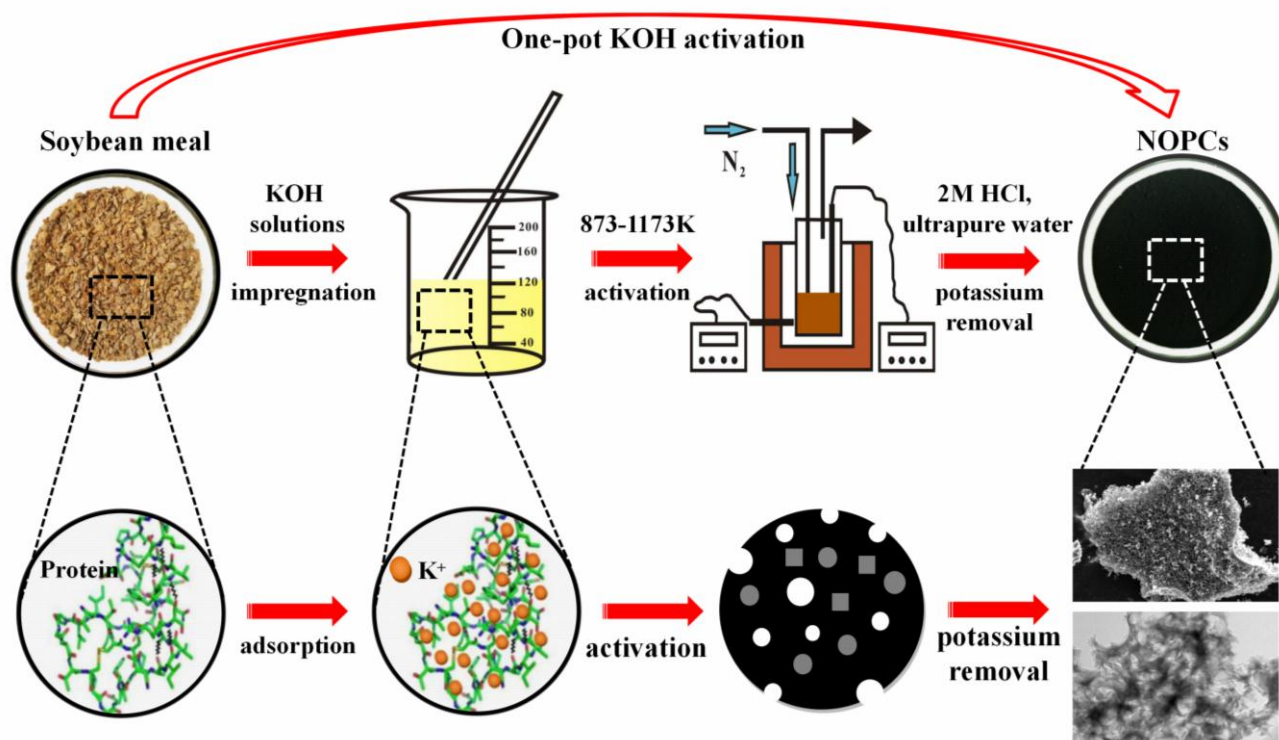
^a Specific surface area, calculated by the Brunauer-Emmett-Teller (BET) method.

^b Total pore volume, measured at $P/P_0 = 0.99$.

^c Micropore volume, estimated from the t-plot method.

^d Mesopore volume, calculated by the difference between total pore volume and micropore volume.

^e average pore size



Scheme 1.

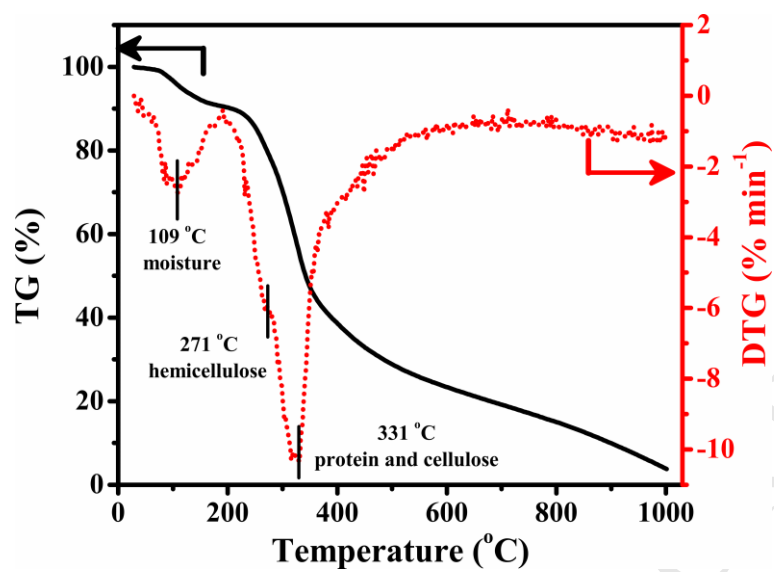


Fig. 1.

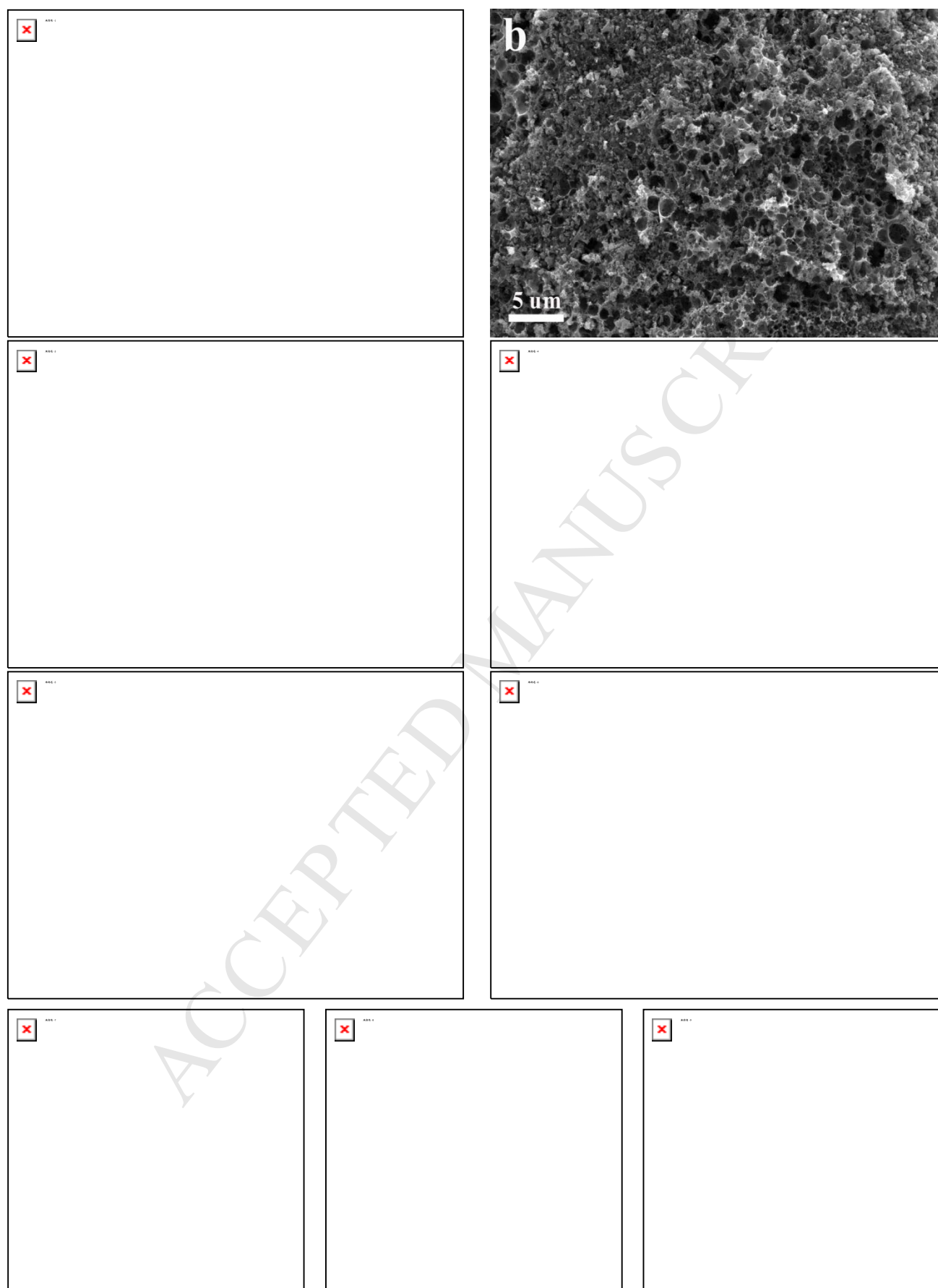
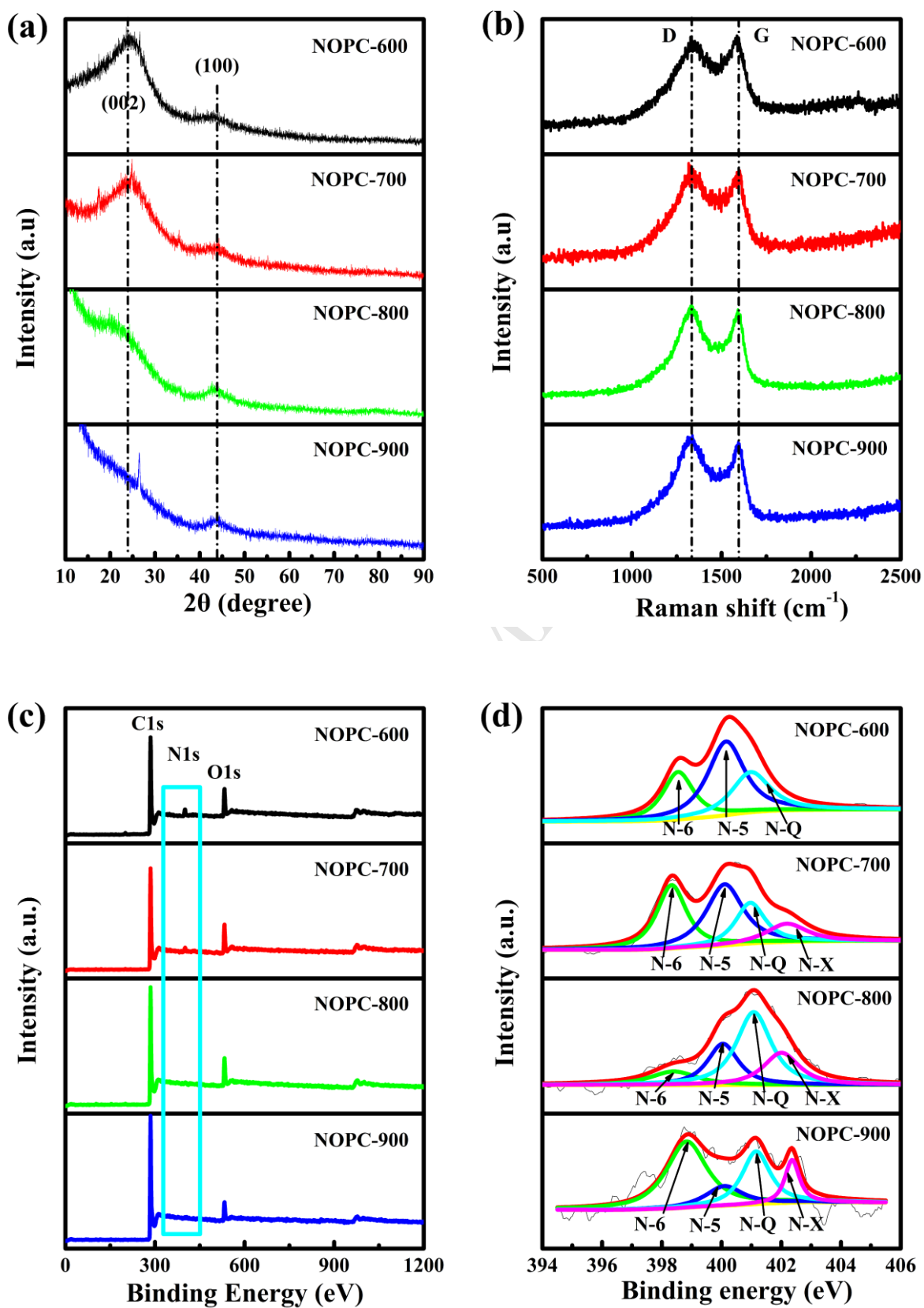


Fig. 2.



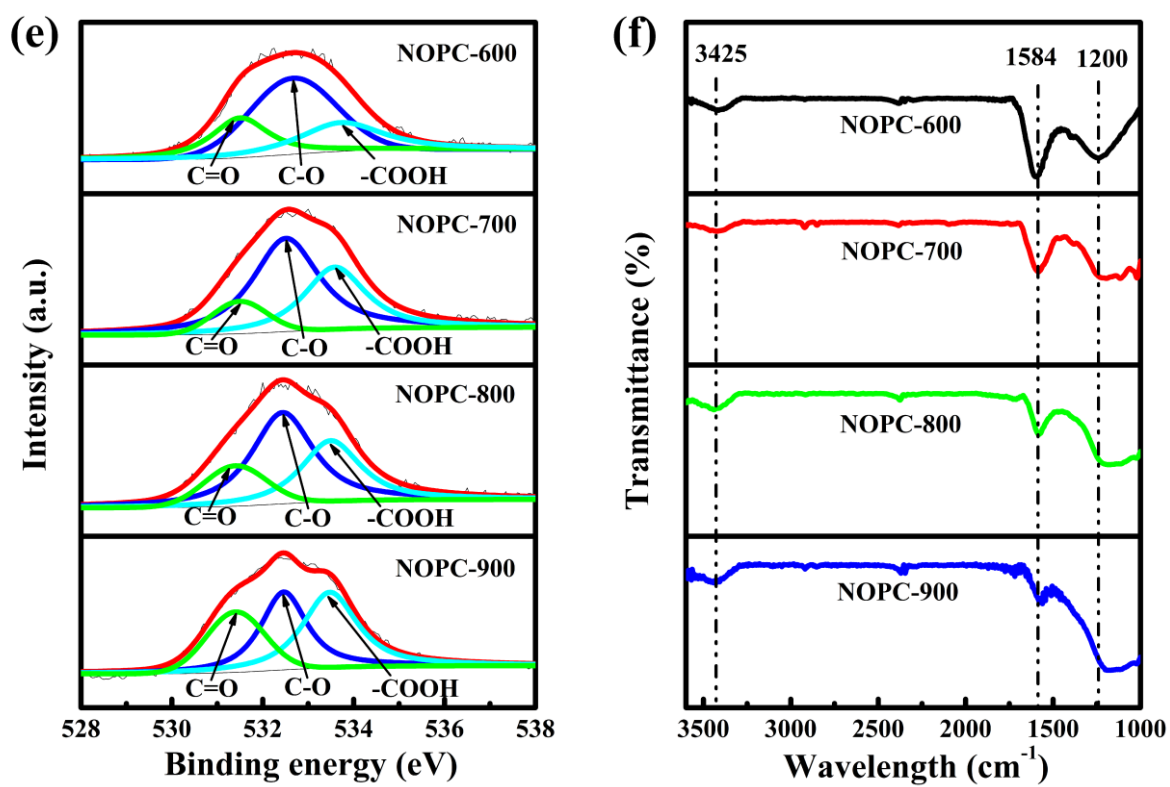


Fig. 3.

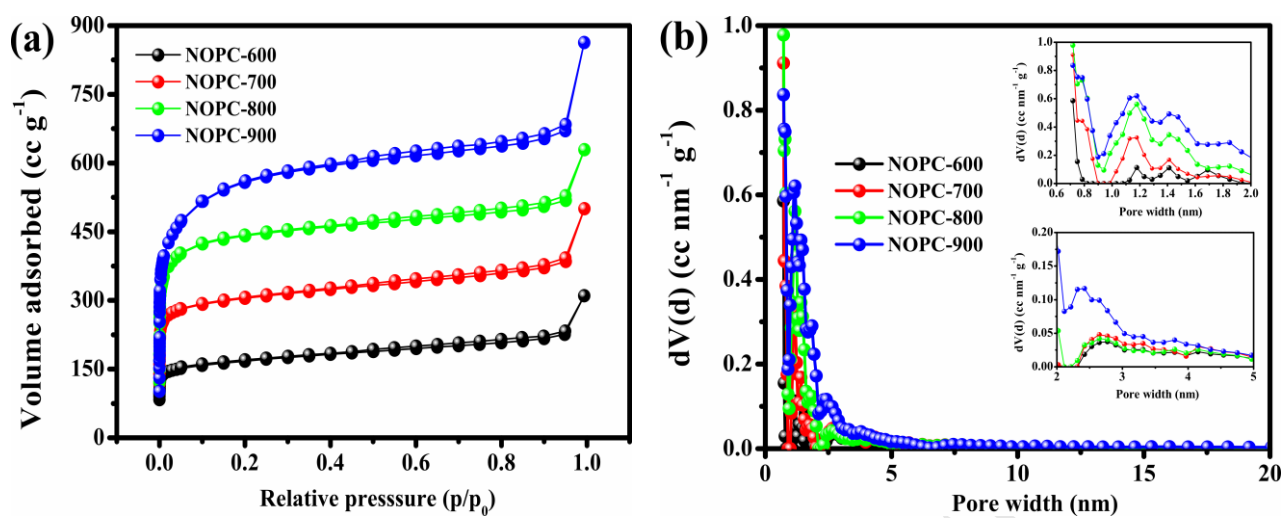


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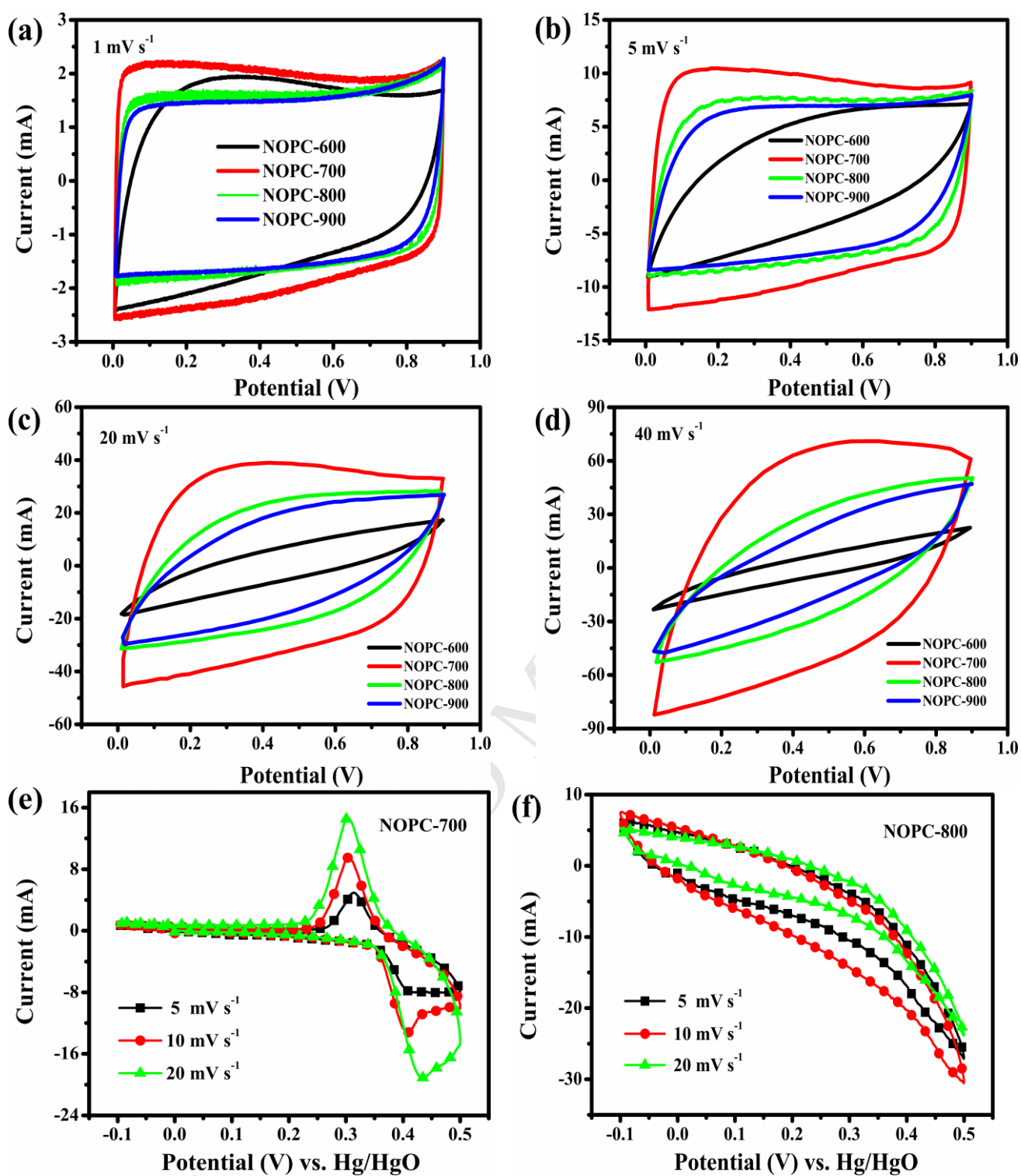


Fig. 5.

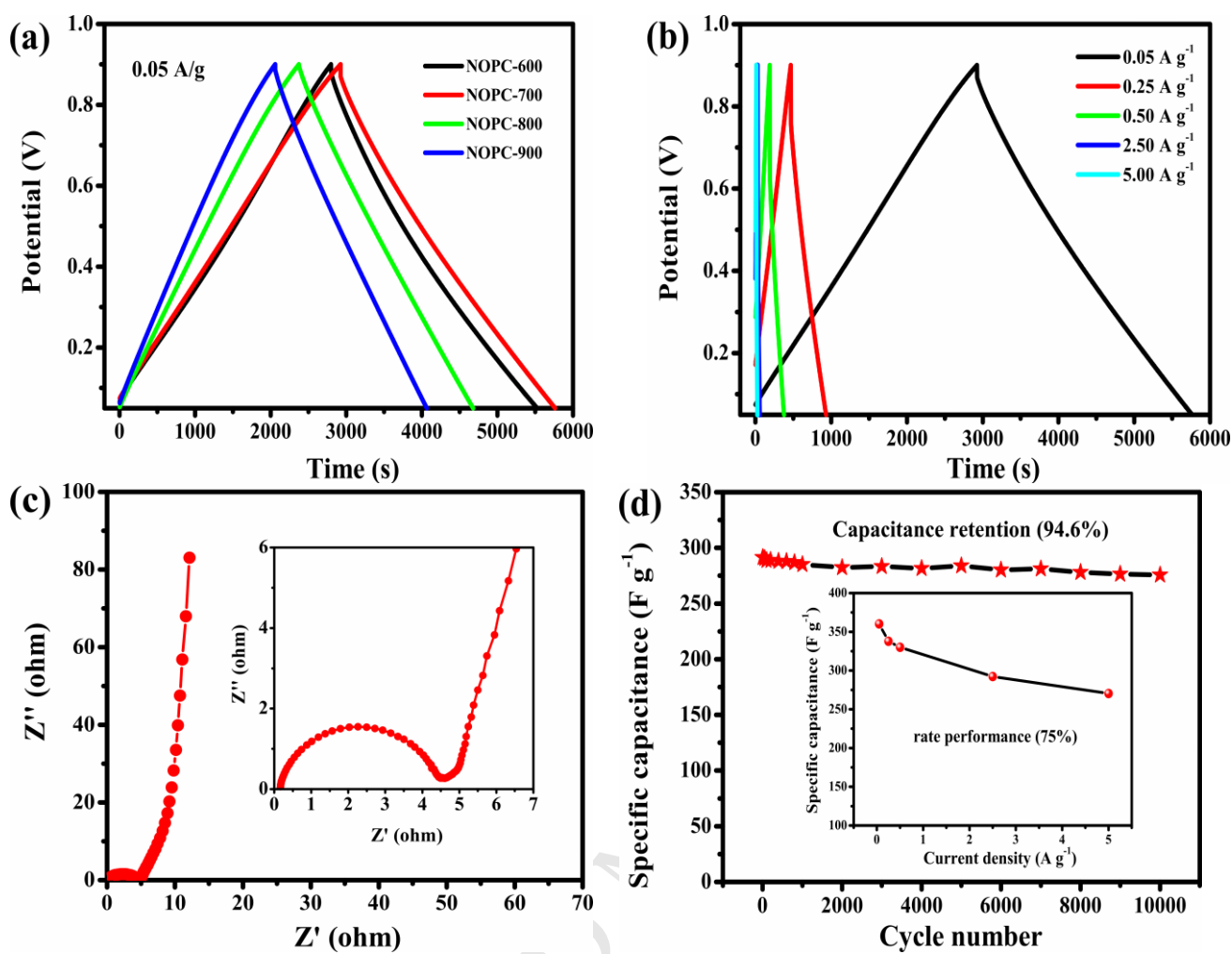


Fig. 6.

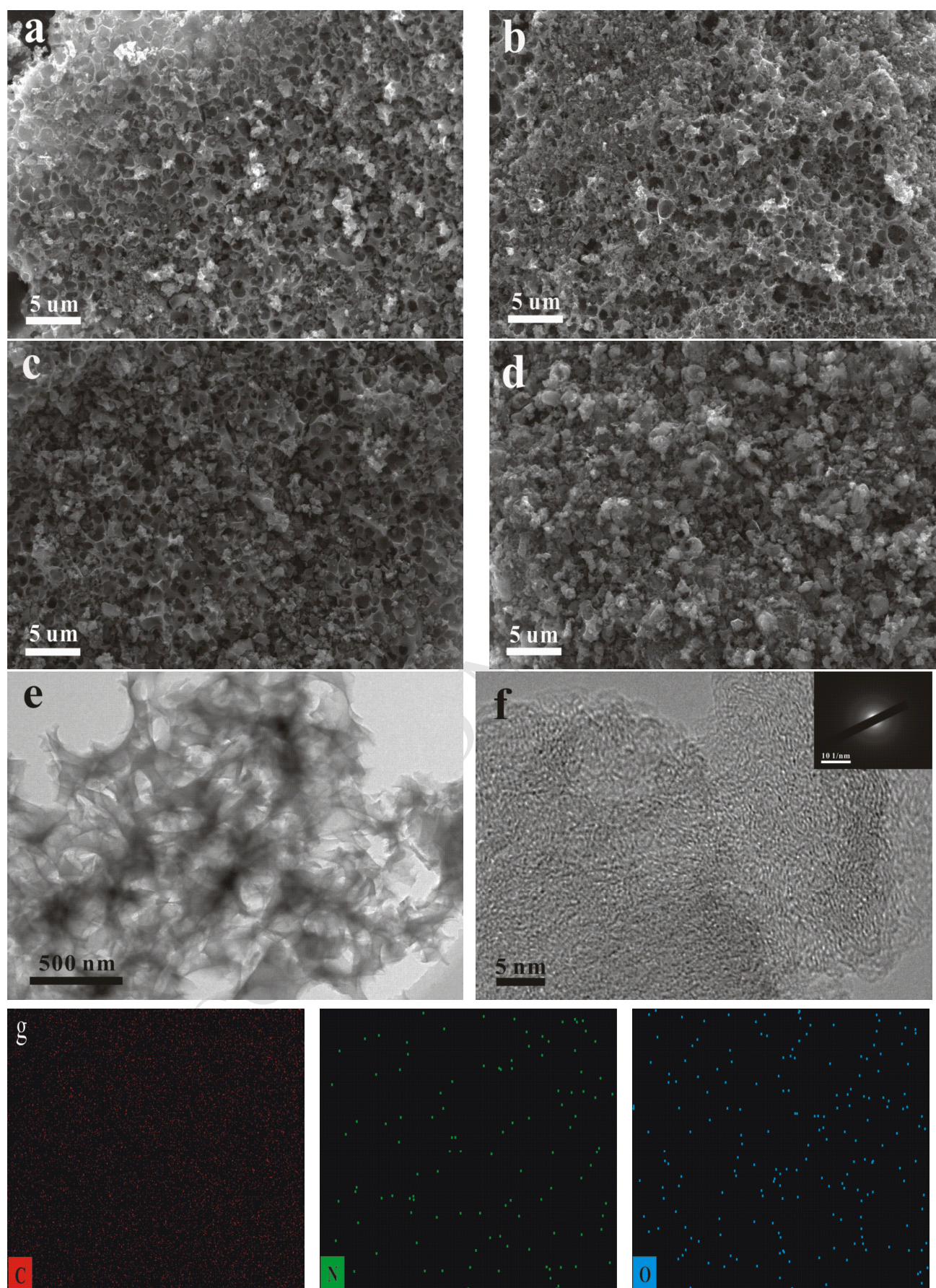


Fig. 2.

Highlights

- Soybean meal possesses the swelling feature when impregnated in KOH solution.
- N and O co-doped porous carbon (NOPCs) were prepared by one-pot KOH activation.
- NOPCs show high specific surface area and well-designed hierarchical porosity.
- NOPCs have uniform nitrogen and oxygen functional groups in framework.
- Supercapacitors with NOPCs as electrodes show superior capacitive behaviors.